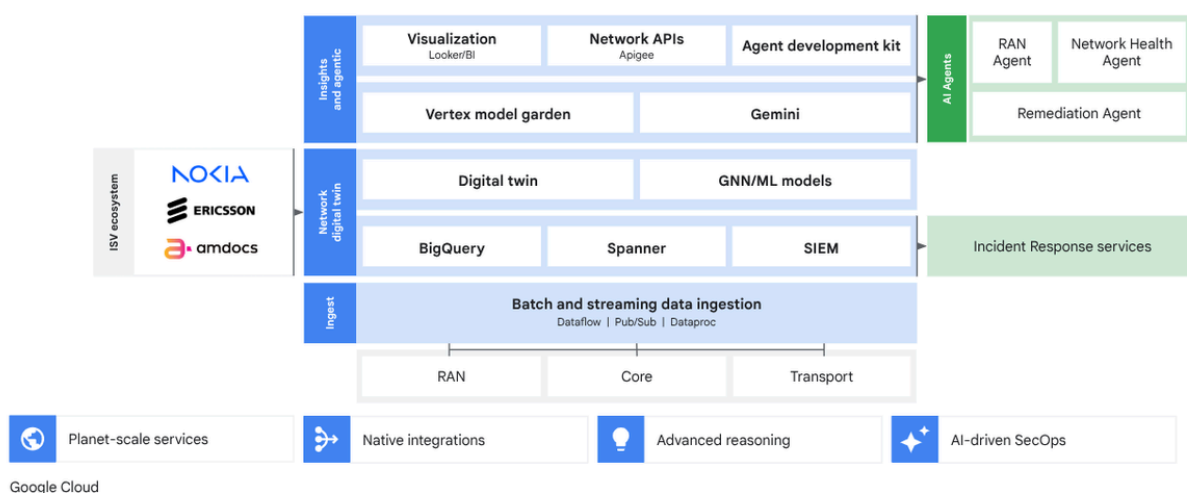


One of the key challenges listed in both the slides and in the paper [The Compute Continuum: Trends and Challenges](#) has been the consideration of the type of framework that a multi-continuum system will take shape in the near future. Since there is no unified framework, corporations and academia alike are rushing out what they think is either the best or the most suitable framework relative to their size and resources.

The example figure in the slides showcases a sample model for the IoT/Industry 4.0-5.0 environment but this is becoming prevalent in other industries as well. Most recently, the telecom industry has demonstrated their preparation towards a move into multi-continuum computing with regards to their operations. Google announces their Autonomous Network Operations Framework, which intends to achieve true network autonomy (<https://cloud.google.com/blog/topics/telecommunications/the-autonomous-network-operations-framework-for-csps>)

Announcing Google Cloud's Autonomous Network Operations framework



As one can see from the provided figure, traditional telecom service providers (like Nokia and Ericsson) have transitioned into more of a SaaS model, running on Google Cloud as its vendor. Since it is a huge financial and logistical burden to compete and invest in billions that major AI vendors can do, it makes sense to focus on the software that directly interacts and integrates into these systems instead of trying to train up your own model.

However, this might also just be a temporary solution. Google Gemini is a generally trained model that is not specific to telecom networks. This translates to a need to develop RAG agents built on Google Cloud's Vertex AI, focusing on the applications in this domain. Due to advancements in computing capabilities, it is not out of the question that a telecom provider can train its own network using a cheaper model that specializes in this type of autonomous network

operations. It remains to be seen if the future multi-continuum model of tomorrow will be built as a single entity or as a cooperation between aligned entities.